Environment: time of day, weather, snow, fog, draw distance – Pilotwings 64 (USA)

Research of 23 October 2026; section 6: what Step 39 built; section 7: the times of day the islands lacked (Step 40). Names from the decompilation (https://github.com/gcsmith/Pilotwings64Decomp, MIT license): environment.c, snow.c, kernel/env.c, kernel/graphics.c, camera.c, skydiving.c, code_72910.c, kernel/code_8170.c. Mission values read from the retail ROM with the editor.

1. Per mission: COMM bytes 8–11 and 0x10–0x2B

COMM	Meaning
byte 8	**Condition** (time of day / weather) 0–5, `taskGet_80345CB0()[0]`
byte 9	**Snow** (1 = on), `[1]`
bytes 10, 11	0 in every mission, not read by the environment code
0x10 f32[3]	base wind vector (x, y, z)
0x1C f32	wind speed range (±)
0x20 f32	rate of the speed's random walk
0x24 f32	wind direction range (degrees, ±)
0x28 f32	rate of the direction's random walk (degrees)

The wind values feed env_802E14E8/env_802E15F0 (random walk every frame, demoRandF). Only Rocket Belt Pilot 2 uses the gusts (±30°, rate 20). env_802E1754 adds a push-back wind beyond 2050 m outside the terrain's bounding box (up to 50 per 2050 m) and the wind of wind objects and thermals.

2. Conditions and environments

`envGetCurrentId` (map, condition) → environment id 2–21; a combination that does not exist gives 22 (the plain "no sky" environment, with the debug message "bad environment specification").

Condition	Holiday	Crescent	Little States	Ever-Frost	Sky model / textures	Fog colour	Colour tint
0 clear day, cumulus clouds	2	7	12	18	0x160: 0x1C5 horizon clouds, 0x1C6–0x1C8 clouds	D1E5FF	–
1 day, altocumulus ("mackerel sky")	3	8	13	19	0x162: 0x1C9	D1E5FF	–
2 overcast	4	9	14	20	0x163: 0x1CA	374242	HSV: S −50 %, V −20 %
3 dark overcast	–	10	15	–	0x167: 0x1CA	1E2D2D / 0F1616	HSV as 2
4 sunset	5	–	16	–	0x164: 0x1C6, 0x1CB, 0x1CD	by heading (below)	RGB ×(0.75, 0.4, 0.2) blend 0.3, HSV
5 night	6	11	17	21	0x165: 0x1C4 stars, 0x1CE moon	000000	RGB ×(0.7, 0.7, 1.0) (EF 0.4, 0.4, 1.0), HSV

- Each environment also draws an island model without fog (0x161 Holiday, 0x166 Crescent, 0x168 Little States, 0x169 Ever-Frost; not identified yet, possibly the far sea/horizon). Texture 0x1CC (clouds) is used by no environment.
- Data: one UVEN file with 24 COMM chunks (env 0–23; limit 32 per level). Each: u8 model count, per model u16 id + u8 flags (8 = follows the camera horizontally, 4 = fogged, 2 = own projection, 1 = keeps state bit 0x200000), then 0x3C bytes ParsedUVEN: clear colour (+0), fog colour (+4), unused colour (+8), fogMin/fogMax f32 (+0x14/+0x18; 998/1000 in all), fogEnabled (+0x1C), clearEnabled (+0x2E). UVLV entries 0x72–0x86 load environment, sky models and textures per environment (`env_802E1A80`, `uvLevelAppend(0x70 + env)`).
- Hard-coded in code: the tints (`env_802E1C1C`: channel 0 = RGB multiply, func_8020F99C/func_8020F630; channel 1 = HSV change, func_8020F5A4 → func_8020B4AC; applied to vertices and textures whose material weight is non-zero) and the terrain palettes (`envLoadTerrainPal`: night palettes 0/1/2 for Holiday/Crescent/Little States, Ever-Frost palette 5 (cond 0/1), 4 (cond 2), 3 (night)).

- Sunset (env 5 and 16 only): `env_802E2060` sets the fog colour each frame from the camera heading, a table of 9 colours from −180° to 180° (grey-blue 46525D away from the sun, warm 7B6252/7D6A55 between 40° and 140°).
- Retail use: conditions 0 ×18, 1 ×14, 2 ×11, 3 ×2, 4 ×6, 5 ×10 of the 61 missions (Cannonball levels count per target).
- Nothing changes during a flight: no day-night cycle, no rain, no lightning.
- **Sky Diving** ignores the mission's condition below the clouds: the free fall uses terrain 9 and environment 1, `skydivingBelow` then always loads condition 1 of the island (env 3, 8, 13, 19). Level 2 (condition 2) is therefore never overcast below the clouds.
- **Hidden night switch** (`env_802E23E0`, every frame): on Holiday Island (terrain 0) within 6 units of (−82, 250, 51) the environment becomes 6 (night) once, with sound 0x6F and a level reload.
- Menus: level select and file menu env 2, title / map / test summary env 23 (0x17), pilot select none.

3. Snow (snow.c)

- On when COMM byte 9 = 1 (code_72910.c), on any island; retail: the 4 overcast Ever-Frost missions (Rocket Belt Class B 2, Gyrocopter Pilot 1, Cannonball level 3 target 3, Birdman stage 2 test 3).
- 800 flakes in a 200-unit cube around a point 100 units ahead of the camera; they fall 15 units/s, drift with the wind and against the camera movement (×3), wrap around the cube.
- Drawn as single white pixels (0xFFFF) straight into the finished 320×240 frame buffer by a graphics callback (uvGfxSetCallback) – no depth test, one pixel at every distance. Emulators that do not copy the frame buffer back (HLE, e.g. Rice) do not show it (same as the module's on-screen text).
- No new flakes inside two boxes (x 310–350.6 / y 34–62.6 / z 151–167 and x 240–344.6 / y 123–154.2 / z 121.5–159.3, Ever-Frost coordinates; probably caves/tunnels) or while the player flag +8 is set.
- Switched off in photo mode, results and replay screens. Memory 0x352C bytes (heap).

4. Fog and draw distance

- Camera: near 1, **far 2000** for every vehicle (camera.c; Cannonball +100 in the cannon view; photo mode changes it for the shot).
- Fog: `uvGfxSetFogFactor(f)` → `gSPFogPosition(f × 1000, 1000)`, capped at 0.996. In flight the environment callback `env_802E0CF0` (uvEnvFunc) sets 0.996 every frame, so the fog starts at about 220 m and reaches about 60 % at 500 m, 85 % at 1 km and 100 % at 2 km (perspective depth; values computed, not measured). Fog colour = the environment's.
- The same callback draws a horizon band in the fog colour (opaque at the far line, 87.5 % of the far distance, transparent towards the camera) to hide the terrain's edge.
- uvEnvProps/uvEnvGetProps (kernel) set fog colour, clear colour and fog factor at run

time; uvEnvGetProps returns the red value for all four clear-colour channels (bug, unused).

- Other limits: at most 64 visible terrain tiles per frame, 4200 graphics commands per frame (Bug list); models switch LODs by their own distance tables (UVMD).

5. What an editor could offer (not decided)

Without code changes: condition (only the combinations of the island), snow, wind and gusts per mission. With the game module: own fog distance and colour per mission (hook after env_802E0CF0), own far distance (watch the 64-tile and 4200-command limits), snow everywhere with density/speed, condition below the clouds for Sky Diving. With new data: the missing combinations (Holiday 3, Crescent 4, Ever-Frost 3 and 4) as new UVEN chunks and UVLV entries plus a patched `envGetCurrentId`; own tints and palettes need patched tables.

6. Built in Step 39: Missions tab, section "Weather and view"

- **Mission file (no code):** time of day / weather (COMM byte 8; since Step 40 every condition on every island, section 7), snow on/off (byte 9; also on for rain), wind as direction (where it blows to, 0° = north = +y, 90° = east = +x) and strength, the variation of direction (± degrees, rate) and of strength (± units/s, rate) (COMM +0x10–0x2B). A Cannonball level edits the weather per target.
- **Settings (project.json, `MissionSettings.Weather`):** rain instead of snow, fall speed (0–30, default snow 15, rain 25), amount (200–1200, default 800), effect colour, fog colour (27 colours: the 6 of the game and 21 more; locked for a sunset), "tint the sky too" (also with the environment's own colour, which is then written into the entry), view distance (1000–2100 m, 50 m steps, default 2000), fog start and density at the view distance (only with "fog follows the view distance" switched off; it is on by default).
- **Data area entry** +0x80 (see DataArea.cs): flags (1 fog colour, 2 sky tint, 4 view distance, 8 fog, 16 snow/rain), kind, colour (RGBA 5551), fog r/g/b, view distance, gSPFogPosition multiplier and offset, fall speed, amount. Header +0x60: the view distance cameraInit loads (2000 in every image).
- **Game module (weather.c, built with the FPU; module.c stays soft-float):**
  - taskLoad's call of func_802EB598 (ROM 0xCC818) → module_env_start: fog colour of the environment (and with tint: the sky dome's vertex colours and the clear colour, multiplied by the fog colour brightened to 255), view distance (DataArea +0x60 for cameraInit, the difference for a camera that is not set up again, so Cannonball's +100 stays), fog command. The last mission's changes are undone first where the data was not loaded anew. taskDeinitLevel's call of func_802EB5E4 (ROM 0xCC8C4) → module_env_end (fog command off, view distance back to 2000).
  - cameraInit (ROM 0x5A6D0/0x5A6D8): "lui at, 0x44FA … mtc1 at, f4" → "lui at, 0x801F … lwc1 f4, 0x5060(at)".
  - uvGfxSetFogFactor (kernel ROM 0x21C50) → module_fog_factor: with the mission's fog the G_MW_FOG word gets its multiplier and offset (fog linear in

depth from the start, the density reached at the view distance; the game's command cannot be steeper than multiplier 32767); otherwise the original (a trampoline runs its first two instructions).
    - snowUpdate (0xC74E4) and snowDraw (0xC7460) → module_effect_update/draw: snow as the game (same random numbers, same projection; VM: 90 of 90 frames pixel for pixel equal without settings) or with the mission's speed, amount and colour; rain as vertical streaks (0.15 s of the fall, at least 3 and at most 24 pixels), half the drop's colour over the picture; rain falls straight down (no drift with the wind or against the camera's movement, unlike the snow). The snow-free boxes only on Ever-Frost Island (terrain 7). The frame callback keeps the module's on-screen messages in front.
    - Sky Diving: skydivingBelow (ROM 0xB9128) calls module_sd_below, which loads the mission's environment (envGetCurrentId, uvLevelAppend(0x70 + env)) instead of condition 1 of the island; built into every hack.
    - The patches (except Sky Diving) are only built in when a mission has weather settings in the data area.
- **Checked:** VM: snow identical to the game; rain streaks; mission start sets fog colour, sky and clear colour tint, view distance (with Cannonball's +100) and the fog command, the next mission without settings gets all back; the fog hook writes the mission's command and the original one after the mission. Emulator (mupen64plus, Rice): night and overcast environments, the sky tinted red, the view distance in RAM (1500 m), the rain running in the real game (375 streaks, 3–28 px). Rice shows neither snow and rain (frame buffer) nor fog reliably: to check in ares.

7. Built in Step 40: every time of day on every island

- **Data:** the four missing combinations become environments 24–27, appended behind the game's 24 chunks of the UVEN file (room for 32), each with its level 0x70 + number appended behind the 136 COMM chunks of the UVLV file (0x88–0x8B), so no number moves. Each is a copy of a template of the same condition with the island's far model swapped:

Env	Island, condition	Template (sky, fog, tint)	Models	Terrain palette of
24	Holiday, dark overcast	10 (Crescent; fog 1E2D2D)	0x167 sky, 0x161	4 (none)
25	Crescent, sunset	5 (Holiday)	0x164 sky, 0x166	7 (none)
26	Ever-Frost, dark overcast	10	0x167 sky, 0x169	20 (palette 4)
27	Ever-Frost, sunset	5	0x164 sky, 0x169	18 (palette 5)

The level chunk is the template's level with the environment number and the island model replaced (lists: terrains, lights, environments, models, contours, textures, sequences, animations, fonts, blits; ParsedUVLV).

- **Island table** (data area header +0x80, its address at header +0x20; `WeatherEnvironments.Table`): 8 islands × (map number of D_80362690->map, the environment of each of the 6 conditions or 0xFF, 0), then per new environment 24–31 (tint template, palette template, 0, 0). Room for future islands and environments.
- **Game module (weather.c):** envGetCurrentId (ROM 0x687E4) jumps to module_env_id (table by map and condition, 22 if none); env_802E1444's calls of env_802E1A80 (0x689B4) and env_802E1C1C (0x689BC) go to module_env_level (uvLevelAppend(0x70 + env) for a new one) and module_env_tint (the template's tint); envLoadTerrainPal (0x68EC0) jumps to module_env_palette (the template's palette, then the original through a trampoline). The sunset's fog by heading (env_802E2060, 0x695A8) tests `mask << env < 0` (bit 31 − env for 5, 16, 25, 27) instead of `env == 5 || env == 16`.
- Everything is built in only when a mission (any Cannonball target too) uses one of the four combinations; Sky Diving below the clouds follows through envGetCurrentId. The import does not take the extended UVEN and UVLV files over as replacement files.
- **Checked:** emulator (mupen64plus, Rice), Hang Glider Beginner on each island through the Classes tab: Holiday dark overcast (env 24, models 0x161/0x167, fog 1E2D2D in RAM), Crescent sunset (25, 0x164/0x166), Ever-Frost dark overcast and sunset (26/27) load and fly; the screenshots show the darker sky and terrain and the sunset tint on the snow. The sunset's fog by heading works through the patched test: Crescent sunset (25) and Holiday sunset (5) both have fog 45515C at heading −90° and 46525D at −150° (RAM of save states). Not checked: in ares (LLE) and by eye in a real flight towards the sun.